SciMON 🧪: Scientific Inspiration Machines Optimized for Novelty

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Abstract

We explore and enhance the ability of neural language models to generate novel scientific directions grounded in literature. Work on literature-based hypothesis generation has traditionally focused on binary link prediction—severely limiting the expressivity of hypotheses. This line of work also does not focus on optimizing novelty. We take a dramatic departure with a novel setting in which models use as input background contexts (e.g., problems, experimental settings, goals), and output *natural language ideas* grounded in literature. We present SciMON, a modeling framework that uses retrieval of "inspirations" from past scientific papers, and explicitly optimizes for novelty by iteratively comparing to prior papers and updating idea suggestions until sufficient novelty is achieved. Comprehensive evaluations reveal that GPT-4 tends to generate ideas with overall low technical depth and novelty, while our methods partially mitigate this issue. Our work represents a first step toward evaluating and developing language models that generate new ideas derived from the scientific literature[1].

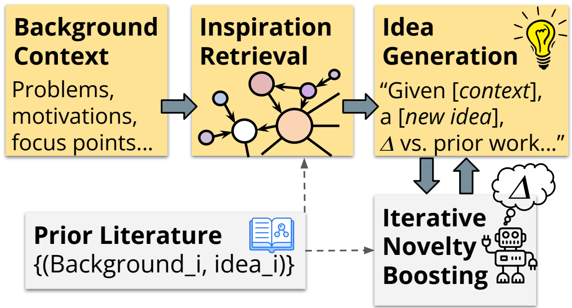

Figure 1: SciMON takes background context and generates ideas grounded in literature inspirations, optimizing novelty by iteratively comparing to related work.

1 Introduction

Can machines mine scientific papers and learn to suggest new directions? The idea that information from the literature can be used for automatically generating hypotheses has been around for decades (Swanson, 1986). To date, the focus has been on a specific setting: hypothesizing links between pairs of concepts (often in drug discovery applications (Henry and McInnes, 2017), e.g., new drug-disease links), where concepts are obtained from papers or knowledge bases previously derived from papers (Sybrandt et al., 2020; Nadkarni et al., 2021).

This common setting has fundamental drawbacks. Reducing the "language of scientific ideas" (Hope et al., 2023) to this simplistic form limits the expressivity of the hypotheses we can hope to generate, and does not capture nuanced *contexts* that scientists consider: target application settings, requirements and constraints, motivations and challenges. In light of the strong progress recently made with large language models (LLMs), in this paper we explore a dramatically different setting: models that take descriptions of problem contexts—and return *natural language* suggestions of novel scientific directions that are grounded in literature.

We develop a framework named SciMON (Scientific Inspiration Machines with Optimization for Novelty), named after Nobel laureate and AI pioneer Herbert Simon who authored early foundational work on automated scientific discovery (Newell and Simon, 1956; Simon, 1973). We first present an automated data collection methodology that collects examples of past problems and proposed ideas from scientific papers. We then use this data for both fine-tuning and in-context training of LLMs—training them to take problem descriptions and output proposed ideas to address them. We observe that state-of-art LLMs (e.g., GPT-4 (OpenAI, 2023)) struggle with generating novel scientific ideas, and contribute a new modeling framework for generating hypotheses that makes progress in improving the hypothesis generation

[1] Code, data, and resources are publicly available for research purposes: https://github.com/eaglew/clbd.

ability of LLMs (Figure 1). Given a background problem description, models first dynamically retrieve *inspirations* from past literature in the form of related problems and their solutions along with contexts from a scientific knowledge graph. These retrieved inspirations serve to ground the generated ideas in existing literature. We then endow models with the ability to iteratively boost the *novelty* of generated ideas. Given an idea $\mathcal{I}$ generated by the LLM at step t, the model compares $\mathcal{I}$ with existing research in the literature; if it finds strongly overlapping research, the model is tasked with updating its idea to be more novel relative to prior work (much like a good researcher would do). We also introduce an *in-context contrastive model* which encourages novelty with respect to background context.

We perform the first comprehensive evaluation of language models for generating scientific ideas in our new generative, contextual setting. We focus on AI/NLP ideas to facilitate analysis by AI researchers themselves, and also demonstrate generalization to the biomedical domain. We design extensive evaluation experiments using human annotators with domain expertise to assess relevance, utility, novelty, and technical depth. Our methods substantially improve the ability of LLMs in our task; however, analyses show that ideas still fall far behind scientific papers in terms of novelty, depth and utility—raising fundamental challenges toward building models that generate scientific ideas.

2 Background and New Setting

We begin with a brief description of related work and background. We then present our novel setting.

Literature-based discovery Nearly four decades have passed since Don Swanson pioneered Literature-Based Discovery (LBD), based on the premise that the literature can be used for generating hypotheses (Swanson, 1986). LBD has been focused on a very specific, narrow type of hypothesis: links between pairs of concepts (often drugs/diseases). The classic formalization of LBD goes back to Swanson (1986) who proposed the "ABC" model where two concepts (terms) A and C are hypothesized as linked if they both co-occur with some intermediate concept B in papers. More recent work has used word vectors (Tshitoyan et al., 2019) or link prediction models (Wang et al., 2019; Sybrandt et al., 2020; Xu et al., 2023) to discover scientific hypotheses as pairwise links between concepts. A tightly related body of research focuses on *scientific knowledge graph link prediction* (Nadkarni et al., 2021), where predicted links may correspond to new hypotheses, and knowledge bases are reflections of existing scientific knowledge in specific domains, derived from literature. A fundamental gap in this line of work is in the lack of approaches for modeling nuanced contexts (Sosa and Altman, 2022) (e.g., the specific settings in which a drug may be relevant for a disease) for generating ideas in open-ended problem settings with unbounded hypothesis spaces, and for optimizing novelty. Our setting can be viewed as a radical departure addressing the limitations in existing settings.

LLMs for Scientific Innovation Large language models (LLMs) have made remarkable progress in interpreting and producing natural language content and handling knowledge-intensive tasks such as in the medical domain (Nori et al., 2023). Very recent work (Boiko et al., 2023) has explored the use of LLMs in a robotic chemistry lab setting, planning chemical syntheses of known compounds and executing experiments. Robotic lab settings are inherently limited to narrow sub-areas where such experiments are possible and relevant. Other very recent work (Huang et al., 2023) used LLMs to produce code for machine learning tasks such as Kaggle competitions, finding that a GPT-4 agent achieved 0% accuracy on research challenges such as BabyLM (Warstadt et al., 2023). GPT-4 has been anecdotally reported as having "strengths less like those of having a human co-author, and more like a mathematician working with a calculator" (Carlini, 2023). Our goal is to conduct a non-anecdotal evaluation and enhancement of strong LLMs' ability to generate novel open-ended scientific ideas.

2.1 SciMON Problem Setting

We are motivated by imagining an AI-based assistant that suggests ideas in natural language. The assistant takes as input background context $\mathcal{B}$ consisting of (1) current problems, motivations, experimental settings and constraints, denoted as $\mathcal{M}$; and optionally (2) a seed term v that should be a focus point of the generated idea $\mathcal{I}$. The seed term is motivated by considering a user-provided cue for the model to limit its hypothesis space. Importantly, generated ideas should not merely paraphrase the background—the output should be *novel* with respect to $\mathcal{B}$ and the broader literature corpus. Figure 2 illustrates the setting, showing a background

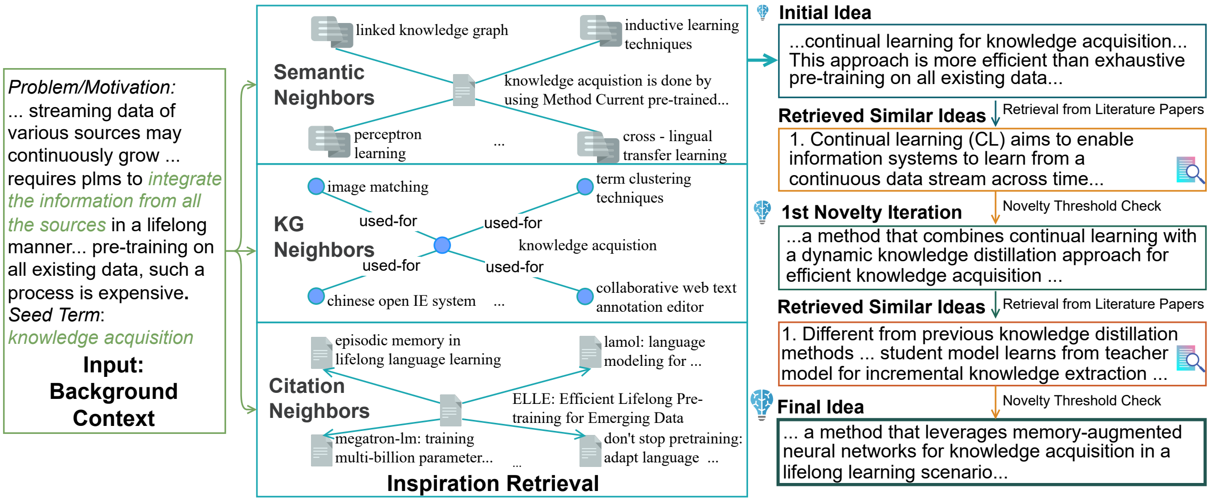

Figure 2: Architecture overview. Our models retrieve *inspirations* and then pass the background input and retrieved inspirations to an LM-based generation module, which iteratively optimizes novelty. Input from Qin et al. (2022).

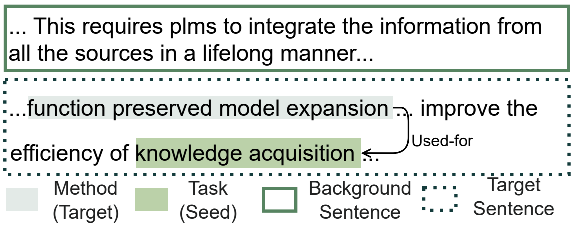

Figure 3: We use IE to obtain literature data for our approach: problems/motivations (background) and proposed ideas (target), as well as salient seed terms.

text that describes problems with *"pretrained language models"* in the lifelong integration of information sources, including computational costs. The assistant aims to generate an idea for performing *"knowledge acquisition"* within this context. Given this input, we aim to generate a full sentence describing a novel idea.

2.2 Automated Training Data Collection

We obtain training data derived from papers with scientific information extraction (IE) models—extracting past examples of background sentences and corresponding ideas (e.g., descriptions of methods used for specific problems in the background sentences), along with salient entities as seed terms. This data is used for training in both in-context learning and fine-tuning setups.

We construct a corpus $\mathcal{D}$ from 67,408 ACL Anthology papers from S2ORC (Lo et al., 2020) (we later also conduct an experiment with a biomedical corpus §4.1). Given a title and the corresponding abstract from a document d, to select problem/motivation sentences $\mathcal{M}$ we first perform scientific sentence classification (Cohan et al., 2019) to classify sentences from the abstract into one of {*Background*, *Method*, *Objective*}, selecting sentences with labels of *Background* and treating the remaining sentences as target sentences $\mathcal{T}$ which will serve as desired output examples (Figure 3).

For seed term selection, we apply a state-of-the-art scientific IE system (Ye et al., 2022) to $\mathcal{T}$ to extract *entities* corresponding to *Task*, *Method*, *Evaluation Metric*, *Material*, and *relations* of the form `[method,used-for,task]`—mentions of methods and the tasks they are used for, materials used for tasks, etc. We treat the head (e.g., method) or tail (e.g., task) entity as the *seed* term, and name the other entity (tail/head, respectively) as a *target* term $t \in \mathcal{T}$. Continuing our example from Figure 2, Figure 3 shows how the seed and target terms ("*knowledge acquisition*" and "*function preserved model expansion*") are extracted from $\mathcal{T}$. During training, each instance contains ($\mathcal{B}$,$\mathcal{T}$) pairs; during evaluation, target information is removed.

We use SciCo (Cattan et al., 2021) to obtain coreference links for entity normalization, and use ScispaCy (Neumann et al., 2019) to replace abbreviations with a more informative long form. We also collect paper metadata, including the citation network $\mathcal{G}_c$. We split our dataset temporally (train/dev/test correspond to papers from years <2021 / 2021 / 2022 respectively). For our experiments, we used model checkpoints trained on data preceding 2022, avoiding the risk of data contamination (§6). Table 1 shows data statistics.[2]

[2]More details are in Appendix C.